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ORIGINAL ARTICLE

Estimating rat predation on Humboldt Penguin colonies in north-central Chile

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Abstract Rats (*Rattus* spp.) are among the most successful alien predators brought to islands by humans and have had devastating impacts on numerous seabird populations, but studies demonstrating rates of consumption and ecological impacts on penguins are scarce and mostly based on anecdotal evidence. We investigated the effects of rat predation on Humboldt Penguins (*Spheniscus humboldti*) by simulating unattended clutches with domestic Chicken eggs. Experiments were independently set at two Humboldt Penguin colonies in north (Pájaros Island, 29°S) and central Chile (Algarrobo Island, 33°S). At both colonies, eggs were primarily predated by rats (*Rattus rattus* = 70.8 % at Pájaros and *Rattus norvegicus* = 52.6 % at Algarrobo), and secondarily by Kelp Gulls (*Larus dominicanus* = 9.7 % at Pájaros and 15.8 % at Algarrobo). Significantly more eggs were predated at night. At both colonies, rates of rat and gull predation were highest within the first 12 h. Our study constitutes the first quantification of rats as important alien predators at Humboldt Penguin colonies. We suggest that rat presence at Humboldt Penguin colonies coupled with events that can cause temporary nest abandonment, such as human perturbation and El Niño events, may impact on the species' breeding success. Eradication of rats is suggested to improve the nesting habitat of this and other threatened and endemic seabird species in the region.

Keywords Introduced species · Island conservation · Nest attendance · Alien predation · *Rattus* · Penguins

Zusammenfassung

Wie sehr räubern Ratten in Kolonien des Humboldt-Pinguins (*Spheniscus humboldti*) in Nord- und Zentral-Chile

Ratten (*Rattus* spp.) gehören zu den erfolgreichsten Räubern, die von Menschen auf Inseln eingeschleppt wurden, und sie hatten bereits verheerende Auswirkungen auf viele Seevögel-Populationen; dennoch liegen nur wenige Untersuchungen über den Grad der Vernichtung und der ökologischen Folgen für Pinguine vor und gehen in der Regel über anekdotische Belege hinaus. Wir untersuchten die Auswirkungen des Räuberns von Ratten unter Humboldt-Pinguinen (*Spheniscus humboldti*), indem wir mit Eiern von Haushühnern unbewachte Gelege vortäuschten. Die Experimente wurden unabhängig voneinander in zwei Kolonien von Humboldt-Pinguinen in Nord- (Pájaro-Inseln, 29°S) und in Zentral-Chile (Algarrobo-Inseln, 33°S) durchgeführt. In beiden Kolonien wurden die Eier in erster Linie von Ratten (*Rattus rattus* = 70.8 %, Pájaro-Inseln, und *Rattus norvegicus* = 52.6 %, Algarrobo-Inseln) und an zweiter Stelle von Dominikanermöwen (*Larus dominicanus* = 9.7 %, Pájaro- und 15.8 %, Algarrobo-Inseln) erbeutet. Nachts wurden deutlich mehr Eier geraubt, und in beiden Kolonien

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erbeuteten sowohl die Ratten, als auch die Möwen die meisten Eier in den ersten 12 Stunden. Unsere Studie ist die erste quantitative Untersuchung zu Ratten als wichtigem eingeschlepptem Räuber in Kolonien von Humboldt-Pinguinen. Das Auftreten von Ratten in Kolonien des Humboldt-Pinguins kann unseres Erachtens in Verbindung mit Ereignissen wie Störungen durch Menschen oder durch Faktoren um El Niño, die das temporäre Verlassen von Gelegen bewirken, Auswirkungen auf den Bruterfolg der Art haben kann. Wir regen die Ausrottung der Ratten dort an, um das Brutgebiet dieser und anderer einheimischer, bedrohter Seevogelarten in der Region zu verbessern.

Introduction

Species introduced by humans either accidentally or deliberately to otherwise inaccessible sites such as islands have caused devastating effects on local bird populations (Moors and Atkinson 1984; Newton 1998; Martin et al. 2000; Jones et al. 2005, 2008; Brooke et al. 2010; Hilton and Cuthbert 2010) and may alter ecosystem functioning on islands (Fukami et al. 2006; Jones 2010). Alien predators have caused numerous bird extinctions during the past 200 years and currently nearly 25 % of all globally threatened birds are affected in some way by introduced animals, especially rats and feral cats (Birdlife International 2000). These birds have evolved in the absence of mammalian predators, thus they often lack effective anti-predator behaviors. Due to their particular (*k*-selected) life history traits, including small clutch size and deferred maturity, seabirds are particularly susceptible to predators introduced into their colonies (Moors and Atkinson 1984; Gaston 2004). Most severely affected are those seabirds that nest in burrows and that leave nests unattended during foraging trips, such as petrels and shearwaters (e.g., Imber 1984; Thibault 1995). Rats (*Rattus* spp.) are among the first and most successful alien predators brought to islands by humans and have had significant impacts on seabird populations (Moors and Atkinson 1984; Jones et al. 2008; Jones 2010). These predators are able to predate on eggs, chicks and even full-grown birds up to or even above their own body weight (Newton 1998; Jones et al. 2008).

Along the extensive Chilean coast in the Southeastern Pacific, both Black (*Rattus rattus*) and Norway Rats (*R. norvegicus*) have a broad distribution along continental and coastal Mediterranean-type habitats (Lobos et al. 2005). At some coastal localities in central and southern Chile, rats have been identified as effective intertidal predators of snails and limpets (Zamorano 1986; Navarrete and Castilla 1993). Rats also occur on several coastal and oceanic islands off Chile where significant seabird colonies exist; several authors have emphasized the potential effects of rats on seabird populations, but most have provided only anecdotic and circumstantial information (e.g., Schlatter 1984, 1987; Simeone and Bernal 2000; Hertel et al. 2005). Recently, Luna-Jorquera et al. (2012) demonstrated that the presence of rats, among others factors, is an important determinant of plant and animal diversity on coastal islands in north-central Chile, although specific effects on seabirds were not assessed by these authors.

Along the Pacific coast of Peru and Chile (between 5° and 42°S), the threatened Humboldt Penguin (*Spheniscus humboldti*) breeds regularly on coastal islands in burrows and rock- and vegetation-covered scrapes (Paredes and Zavalaga 2001) during the austral spring and winter (Paredes et al. 2002; Simeone et al. 2002). Both sexes share parental duties and usually eggs are never left unattended (Williams 1995; Davis and Renner 2003), but parents may temporarily abandon them because of prolonged foraging trips induced by food scarcity during El Niño events (Culik et al. 2000) and human disturbance (Luna-Jorquera 1996; Ellenberg et al. 2006). In such cases, if penguins return to their nests within 1–3 days, incubation may be resumed with no apparent effect on hatching success (as also seen in Magellanic Penguins *Spheniscus magellanicus*; Yorio and Boersma 1994). However, it is not infrequent that eggs disappear by the time the penguins return to resume incubation. Evidence found at or near the nest suggests that some predators are involved in egg loss during nest inattendance and it is conceivable that such predation is affecting the breeding success of Humboldt Penguins. By means of a simple experiment using chicken eggs we aimed to identify predators involved in egg loss and quantify the rate at which eggs disappear from experimental unattended Humboldt Penguin nests at two islands in north-central Chile.

Methods

Our study was conducted at seabird colonies on two islands located off the coast of north-central Chile. Experiments were conducted from 8 to 11 December 2003 at Algarrobo Island (33°21′S, 71°41′W) and from 13 to 19 December 2003 at Pájaros Island (29°35′S, 71°33′W). Algarrobo (also known as Pájaro Niño Island) is a near-shore 3-ha island that was connected to the mainland in 1977 by a breakwater wall near the city of Algarrobo and supports a population of 250 pairs of Humboldt Penguins (Simeone and Bernal 2000; Simeone et al. 2003). Pájaros is a 124-ha island located 40 km north of the city of Coquimbo and 21 km off the coast that supports a population of 600 pairs of Humboldt Penguins (Simeone et al. 2003). Algarrobo is a designated Nature Sanctuary, but Pájaros has no official protection.

At both colonies, potential aerial predators included Kelp Gulls (Algarrobo: 450 pairs; Pájaros: 2,000 pairs; Simeone et al. 2003) and Turkey Vultures (*Cathartes aura*). At Algarrobo, Black Vultures (*Coragyps atratus*) were also present (personal observation). As for potential terrestrial predators, based on field observations and analysis of several skulls of dead animals (identified using Reise 1973), we confirmed the presence of non-native Black Rats on Pájaros Island and Norway Rats on Algarrobo Island. In addition, feral cats (*Felis catus*) have been reported on Algarrobo Island (Simeone and Bernal 2000). At both colonies, rats were probably introduced by fishermen while cats moved onto Algarrobo after the breakwater wall construction (Simeone and Bernal 2000; Simeone et al. 2003). Pájaros Island has been identified as an island with high anthropogenic impact, derived mostly from guano collection, seabird egg extraction and rat presence (Luna-Jorquera et al. 2012).

We conducted the experiments using domestic Chicken (*Gallus domesticus*) eggs to resemble normal Humboldt Penguin clutches. We selected 80 Humboldt Penguin nests (40 at each colony) and placed two chicken eggs in each. Penguin and chicken eggs are both white, but penguin eggs are larger and heavier (73.2 ± 3.0 mm length, 55.2 ± 1.8 mm width, 120.3 ± 9.3 g, $n = 88$) than the chicken eggs (56.1 ± 1.8 mm length, 42.7 ± 0.6 mm width, 54.5 ± 4.1 g, $n = 160$) used in our experiments. To avoid the risk of pathogen transmission into the wild penguin population (e.g., salmonella, campylobacter, mycoplasma, paramyxoviruses, and pox viruses), chicken eggs were boiled for 10 min prior to use and were removed after the experiment ended (if still present). At the time of the study (late in the breeding season for Humboldt Penguins), none of the experimental nests were occupied by breeding penguins, but had been active in previous breeding seasons (personal observation). In this way, we attempted to reduce disturbance to nesting birds.

On both islands, we used rock-covered nests, which is the most abundant nest-type at both colonies (>50 % of total nests; A. Simeone, unpublished data) in order to facilitate inter-island comparisons. Experimental nests were 10–15 m apart and nest floors were carefully cleaned with a brush, and then fine sand was scattered inside the nests in a 20-cm-diameter circle, with the two chicken eggs placed in the center of the sand platform. By inspection of the sand platform and nest periphery, we attempted to identify predators. Predators were divided into three categories (Yorio and Boersma 1994): (1) avian predator, if characteristic bird footprints and/or broken eggs were away from the nest and had a hole characteristic of avian predators, (2) mammalian predator, if characteristic mammal footprints and scats were present in the sand platform and if egg shells were broken in small fragments, and (3) unknown predator, if eggs disappeared from the nest and no clear signs that allowed predator identification were present. We considered eggs predated when broken open, partially eaten, or when missing from nests. Because Humboldt Penguins can leave eggs unattended from 1 to 3 days without affecting egg survival (personal observation; see also Yorio and Boersma 1994), our experiments were designed to last up to 72 h. Nests were checked at 12-h intervals around 0800 and 2000 hours, shortly after dawn and before dusk, respectively. Eggs missing in the morning checks were considered predated in the previous night, while those missing in the evening checks were considered predated during the same day.

To determine if the proportion of predated eggs at the end of the experiments differed between islands, a contingency table was applied. Categories across both colonies were: number of intact eggs (i.e., not predated), number of eggs predated by gulls, number of eggs predated by rats, and unknown predators. So as to estimate the number of eggs predated over time, for each predator at each island, we performed a regression between the cumulative number of predated eggs and time. This allowed us to calculate the cumulative rate of egg predation and to compare it using a test for slopes. For each fitted model, we analyzed the residuals to check for normality and constant variance. To test for possible autocorrelation, we used the Durbin–Watson statistic. To accomplish the regression assumptions, time was transformed to Ln $(x + 1)$. Durbin–Watson values were larger than the upper critical value ($d_U = 1.356$ for $n = 7$ and $k = 1$, obtained from Fry 1993), indicating that no significant autocorrelation was evident. In only one case (Kelp Gulls at Algarrobo) was the Durbin–Watson value = 0.9820, suggesting that independence was uncertain. However, examining the autocorrelation plot for this dataset, we observed that lag values were within the 95 % confidence intervals, indicating that each residual is not predictable from the preceding one. As all the other assumptions were met (for all fitted models), we consider this a minor departure and accept the analysis as valid. All analyses were conducted using SYSTAT v.13.

Results

Of the initial 40 experimental nests at each colony, 2 at Algarrobo and 4 at Pájaros were subsequently occupied by Humboldt Penguins during the experiment and were thus excluded from analysis. The fate of experimental eggs placed in Humboldt Penguin nests is shown in Fig. 1. Predation of unattended eggs was attributed to rats and Kelp Gulls and none to either vultures or feral cats. Overall egg survival significantly differed between the two islands ($\chi^2_{0.05,\ 3} = 8.8$, $P = 0.032$; Fig. 1).

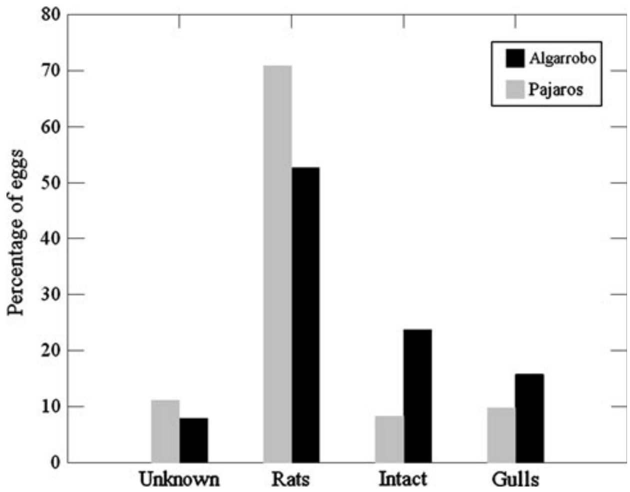

Fig. 1 Fate of experimental eggs placed in Humboldt Penguin (*Spheniscus humboldti*) nests on Pájaros ($n = 72$ eggs) and Algarrobo ($n = 76$ eggs) after 72 h of experiment setting. *Intact* no external damage in egg structure, *rats* predation attributed to rats, *gulls* predation attributed to gulls, *unknown* predator not determined

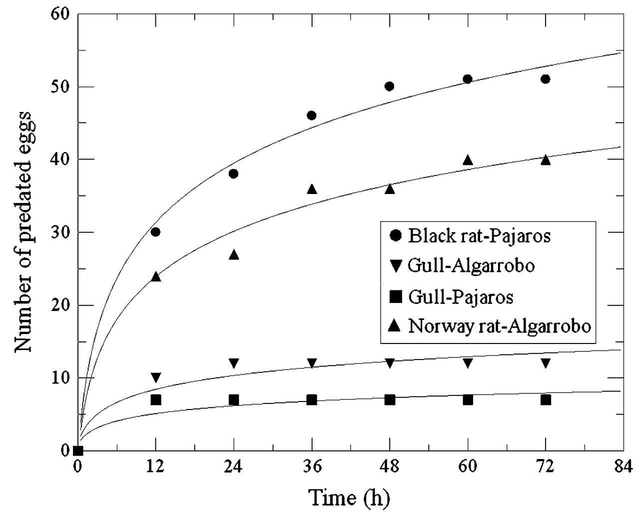

Fig. 2 Rates of egg predation at experimental Humboldt Penguin nests by Kelp Gulls and rats at two colonies in north-central Chile. Values are expressed as cumulative number of predated eggs. The best adjusted models for the equations for each *curve* are provided in Table 2

On Pájaros, rats predated significantly more eggs during the night and gulls during daytime (Fisher's exact test, $P < 0.0003$; Table 1). The same was determined for Algarrobo ($P < 0.001$; Table 1). At both colonies, rates of egg predation by rats and gulls were highest within the first 12 h of the experiment (Fig. 2). After 12 h, rat predation continued (although at a lower rate) and stabilized at 48–60 h. Gull predation rates remained stable after 12 h (i.e., only 1–3 eggs were predated after this time until the end of the experiment).

All comparisons of predation rates shown in Fig. 2 (i.e., for a same predator between islands and for different predators within the same island) were significantly different (test for slopes $F_{1,23} = 7.7$, $P = 0.011$ and Tukey post hoc test, $P < 0.05$). For all the curves shown in Fig. 2, correlation coefficients were >0.92 (Table 2) and highly significant ($P < 0.001$). The parameters for the equations describing each curve in Fig. 2 are given in Table 2.

Table 1 Number of eggs predated by rats and gulls on both islands at different times of the day

	Rats	Gulls
Pájaros		
Day	9	6
Night	42	1
Algarrobo		
Day	3	10
Night	37	2

Discussion

Several experiments have attempted to assess nest predation by using artificial nests (e.g., Leimgruber et al. 1994; VanderWerf 2001; Prieto et al. 2003; Smith et al. 2008). Such studies, however, may provide figures that do not accurately reflect those of real nests, tending either to overestimate (Berry and Lill 2003) or underestimate (Davidson and Bollinger 2000) this important source of mortality. Moreover, the type of artificial nest used can strongly affect the rate at which they are visited by predators (Smith et al. 2008). We overcame this problem by using real Humboldt Penguin nests that were not in use at the time of the experiment, but that we knew were used in previous seasons (personal observation). The fact that some nests were reused by penguins underscores that these nests continued to be suitable for penguins at the time of this study.

Another source of bias in estimating predation is the use of artificial eggs (DeGraaf and Maier 1996; Rangen et al. 2000; Prieto et al. 2003; Smith et al. 2008). We used domestic Chicken eggs as this was the only commercially available and ethical alternative in large quantities that resembled penguin eggs at least in coloration. Since both Norway and Black Rats are considerably larger (320 and 158 g, respectively; Jaksic and Marti 1984) than the chicken eggs used in this experiment, these eggs may have been more often preyed upon because of the ease of moving and carrying them compared to penguin eggs. Nevertheless, Blight et al. (1999) demonstrated that the small native *Peromyscus keeni* (43.8 g) was able to open

Table 2 Parameters used to calculate the cumulative numbers of predated eggs represented by curves in Fig. 2

Island	Predator	Intercept (*a*)	Rate (*b*)	*r*	*P*
Algarrobo	Kelp Gull	1.048	2.889	0.96	0.0006
	Norway Rat	−0.413	9.493	0.99	<0.0001
Pajaros	Kelp Gull	0.924	1.638	0.92	0.0034
	Black Rat	−0.452	12.410	0.99	<0.0001

The model for each case is best described by $Y = a + b \times \text{Ln time}$. Differences between rates were highly significant (test for slopes, $P < 0.001$)

and eat intact Rhinoceros Auklet's (*Cerorhinca monocerata*) eggs which were nearly twice their body mass (79 g). A further aspect to consider is that human and boiled chicken egg odors may have resulted in biased rates of predation, as olfactory-searching predators can cue on them to locate experimental nests and eggs (Whelan et al. 1994; Rangen et al. 2000). Additionally, we cannot ascertain whether boiled eggs produced differential attraction to rats compared to fresh eggs.

Our results allowed us to compare alien (rats) versus native (gulls) predation. Predation by Kelp Gulls (at both islands) compares well with values found by Yorio and Boersma (1994) at Magellanic Penguin colonies in Punta Tombo, Patagonia. These authors reported that native Kelp Gulls and Hairy Armadillos (*Chaetophractus villosus*) combined took on average 6.3 % (range 4–18 %) of unattended penguin eggs. Both rate and magnitude of egg predation were considerably larger in rats compared to gulls, although it is uncertain if gulls could have had a larger impact if rats were absent from the colony. Differences in the rate of predation between gulls and rats may reflect different search strategies. Kelp Gulls are mainly visual predators and locate eggs by aerial searching, especially at those nests with reduced cover (Gandini et al. 1999). Rats, like many other nocturnal mammals, rely mostly on olfactory cues to detect food (Whelan et al. 1994; Rangen et al. 2000), and this could allow them to detect more hidden nests such as those used in this study; this may have also hindered gulls from detecting more eggs from the air. The fact that more eggs were taken at night by rats and more eggs taken during daylight by gulls supports this idea.

The highest rate of egg predation exerted by Kelp Gulls on Algarrobo compared to Pájaros could be reflecting the higher density of gulls at the former colony (150 vs. 16 pairs/ha; Simeone et al. 2003). We have no data on rat abundance on either island to suggest a similar density-related explanation. We do not rule out, however, that the highest rate of egg predation exerted by Black Rats on Pájaros compared to Norway Rats on Algarrobo include species-specific differences in predation strategies (Lobos et al. 2005).

A further aspect to be considered is that Humboldt Penguins breed twice a year (Paredes et al. 2002; Simeone et al. 2002), with a first peak during spring and a second during winter. Our study was conducted during late spring, when other seabirds, e.g., Kelp Gulls, Peruvian Pelicans (*Pelecanus thagus*), and Peruvian Boobies (*Sula variegata*), also nest at these colonies (Simeone et al. 2003) and may thus provide alternative prey (eggs and chicks) for rats. Hertel et al. (2005) observed rat predation on Peruvian Booby eggs at Pájaros 1 (=Pájaros Sur), indicating that this is indeed the case. It is thus conceivable that, during the penguin's winter breeding season, predation pressure by rats on penguins may increase as no other seabirds are nesting at the colonies.

Despite all potential constraints of our experimental set-up and criticism on the use of artificial nests and eggs for estimating predation, we consider this contribution a useful tool (preferable to any lack of knowledge) that should be used with prudence. Our simulations considered temporary nest inattendance such as that observed as the result of human perturbation and El Niño events (see Luna-Jorquera 1996; Culik et al. 2000; Ellenberg et al. 2006). These are increasing situations at colonies in north-central Chile, and their impacts can further worsen where rats are present. Considering the results presented in this study and their potential implications for Humboldt Penguins (and maybe other seabirds), rat eradication is recommended before they become a conservation problem to this threatened and endemic species (see recommendations on rat eradication by IUCN: Araya et al. 1999; Luna-Jorquera et al. 2002), within a region considered of interest for the conservation of biodiversity at a global scale (Olson and Dinerstein 1998).

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